The streamgraph is a sublime piece of information visualization that represents variation and change, like the boxplot. It may seem like a difficult thing to create, until you start to put the pieces together. Ultimately, a streamgraph is what's known as a *stacked chart*. The layers accrete upon each other and adjust the area of the elements above and below, based on the space taken up by the components closer to the center. It appears organic because that accretive nature mimics the way many organisms grow, and seems to imply the kinds of emergent properties that govern the growth and decay of organisms. We'll interpret its appearance later, but first let's figure out how to build it.

The reason we're looking at a streamgraph is because it's not that exotic. A streamgraph is a stacked graph, which means it's fundamentally similar to your earlier line charts. By learning how to make it, you can better understand another kind of generator, `d3.svg.area()`. The first thing you need is data that's amenable to this kind of visualization. Let's follow the *New York Times*, from which we get the streamgraph in figure 4.21, and work with the gross earnings for six movies over the course of nine days. Each datapoint is therefore the amount of money a movie made on a particular day.

**Listing 4.11 movies.csv**

```
day,movie1,movie2,movie3,movie4,movie5,movie6
1,20,8,3,0,0,0
2,18,5,1,13,0,0
3,14,3,1,10,0,0
4,7,3,0,5,27,15
5,4,3,0,2,20,14
6,3,1,0,0,10,13
7,2,0,0,0,8,12
8,0,0,0,0,6,11
9,0,0,0,0,3,9
10,0,0,0,0,1,8
```

To build a streamgraph, you need to get more sophisticated with the way you access data and feed it to generators when drawing lines. In our earlier example, we created three different line generators for our dataset, but that's terribly inefficient. We also used simple functions to draw the lines. But we'll need more than that to draw something like a streamgraph. Even if you think you won't want to draw streamgraphs (and there are reasons why you may not, which we'll get into at the end of this section), the important thing to focus on when you look at listing 4.11 is how you use accessors with D3's line and, later, area generators.

**Listing 4.12 The callback function to draw movies.csv as a line chart**

```
var xScale = d3.scale.linear().domain([ 1, 8 ]).range([ 20, 470 ]);
var yScale = d3.scale.linear().domain([ 0, 100 ]).range([ 480, 20 ]);

for (x in data[0]) {
      if (x != "day") {
```

**Iterates through our data attributes with a `for` loop, where x is the name of each column from our data ("day", "movie1", "movie2", and so on), which allows us to dynamically create and call generators**